

# BUILDING THE HUMAN CIRCULATORY SYSTEM MODEL

Interactive Hands-On Science Lesson Guide & Activity Plan

## 1. Warm-Up Activity: Feel Your Heartbeat!

### Activity: Finding Your Pulse

Place two fingers gently on your wrist or the side of your neck. Can you feel a steady rhythmic thumping?

- **How It Works:** Your heart acts like a continuous, non-stop *water motor* pumping blood throughout your entire body.
- **Pulse Recheck Challenge:** Jump on the spot for 30 seconds! Check your pulse again. Notice how much faster and harder your heart is pumping to deliver energy and oxygen to your working muscles.

## 2. Meet "Raju Red-Cell" & Core Concepts

Follow the journey of **Raju Red-Cell**, a red blood cell traveling through your body in less than a single minute!

| Component / Concept | Biological Role                                         | Real-World Analogy                                              |
|---------------------|---------------------------------------------------------|-----------------------------------------------------------------|
| Red Blood           | Oxygenated blood (full of fresh oxygen from the lungs). | Refueled train carrying essential goods.                        |
| Blue Blood          | Deoxygenated blood (oxygen used up by the body).        | Empty train returning to get refueled.                          |
| Blood Vessels       | Highways and tubes connecting every organ and cell.     | <b>Indian Railways</b> connecting every city across the nation. |
| Lungs               | Exchanges carbon dioxide for fresh oxygen.              | A <b>CNG / Fuel Station</b> refuelling the red blood cells.     |
| Heart Valves        | Flaps that prevent blood from flowing backward.         | <b>Metro Turnstiles</b> that only let passengers pass ONE way.  |

### The Continuous Loop Journey

**Body → Heart → Lungs (Oxygen Refuel) → Heart → Body (Round & Round!)**

## 3. Hands-On Activity: Let's Get Building!

Build your working hands-on model of the human circulatory system using everyday laboratory materials.

### Materials Needed:

- 2 Syringes & Clear flexible tubing
- 2 One-way Check Valves
- Red & Blue food coloring dissolved in water
- Cardboard body silhouette template

- Zip ties or tape, and 2 small cups

### Step-by-Step Construction Guide:

1. **Step 1: Connect It Up** — Attach the clear tubing and check valves to both syringes. *Crucial: Ensure check valves face the correct direction so water flows strictly in one direction!*
2. **Step 2: Add Colour** — Fill one syringe with red water (oxygenated blood) and the other with blue water (deoxygenated blood). Watch the colors move through the tubing as you push!
3. **Step 3: Mount & Display** — Secure your tubing and syringes onto the cardboard silhouette using zip ties or tape. Your circulatory system model is ready to pump!

**Top Tip:** Work in pairs! Ask a partner to hold the cardboard steady while you attach the tubing and check valves.

# STUDENT PRACTICE WORKSHEET

## Human Circulatory System Model — Test Your Knowledge

Instructions: Answer the following questions based on the lesson and model activity.

### Question 1 (Multiple Choice)

What color is used to represent oxygenated blood in medical diagrams and in our model?

A) Blue

B) Red

C) Yellow

D) Green

### Question 2 (Multiple Choice)

Which component in our working model (and in the human heart) prevents blood from flowing backward, behaving like a metro turnstile?

A) Clear tubing

B) Syringe plunger

C) Check valve

D) Food coloring

### Question 3 (Short Answer)

Why does your heart beat faster when you jump or run for 30 seconds compared to when you are resting?

### Question 4 (Analogy Matching)

Match each part of the human body circulatory system to its real-world analogy from the lesson:

1. **Heart** — \_\_\_\_\_

2. **Blood Vessels** — \_\_\_\_\_

3. **Lungs** — \_\_\_\_\_

4. **Valves** — \_\_\_\_\_

### Question 5 (Flow Order Sequence)

Arrange the following stages of Raju Red-Cell's journey in the correct chronological loop:

☐ Blood travels to the body to deliver oxygen

☐ Deoxygenated blood returns to the heart

☐ Blood goes to the lungs to pick up oxygen

☐ Oxygenated blood enters the heart to be pumped out

### Question 6 (Model Reflection)

When building your model, what would happen if the check valve was installed upside down (facing the wrong direction)? Explain how this affects blood flow.